BATTERY STATE OF HEALTH (SOH) PREDICTION USING MULTI-MODAL DEEP LEARNING

PROJECT REPORT

Submitted By:

Priyanshu Kumar
Roll No: 24075071

Mayank Singh
Roll No: 24075055

Under the Guidance of:

Dr. Aparajita Khan
(Assistant Professor, Department of CSE)

Department of Computer Science and Engineering

Indian Institute of Technology (BHU), Varanasi

2025–2026

Certificate

This is to certify that the project report entitled **"Battery State of Health (SOH) and Remaining Useful Life (RUL) Prediction using Multi-Modal Deep Learning"** submitted by **Priyanshu Kumar (Roll No: 24075071)** and **Mayank Singh (Roll No: 24075055)** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at the Indian Institute of Technology (BHU), Varanasi is a bonafide record of the work carried out under my supervision and guidance during the academic year **2025–2026**.

To the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion by this or any other institute.

Guide Signature:	**Head of Department:**
Dr. Aparajita Khan Assistant Professor Department of CSE, IIT (BHU) Varanasi	Dr. Bhaskar Biswas Head of Department Department of CSE, IIT (BHU) Varanasi
Date:______________	Place:______________

Declaration

We hereby declare that the project report entitled **“Battery State of Health (SOH) and Remaining Useful Life (RUL) Prediction using Multi-Modal Deep Learning”** is an original piece of work carried out by us under the guidance of **Dr. Aparajita Khan**, Assistant Professor, Department of Computer Science and Engineering, Indian Institute of Technology (BHU), Varanasi.

This work has not been submitted elsewhere, either in part or in full, for the award of any degree, diploma, or certificate. All sources of information used in the preparation of this report have been duly acknowledged.

Signature of Student:

Priyanshu Kumar
Roll No: 24075071

Date:______________

Signature of Student:

Mayank Singh
Roll No: 24075055

Date:______________

Acknowledgement

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We are deeply thankful to the **CALCE Battery Research Group** at the University of Maryland for making their battery cycling dataset publicly available, which formed the experimental foundation of this work.

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Finally, we are profoundly grateful to our families and friends for their unwavering moral support and motivation throughout the course of this project.

Contents

1. Abstract

Lithium-ion batteries are the cornerstone of modern energy infrastructure, powering electric vehicles (EVs), large-scale energy storage systems (ESS), and portable electronics. However, electrochemical aging, repeated charge–discharge cycling, and environmental stress cause an inevitable and non-linear decline in battery capacity, making accurate, real-time estimation of the **State of Health (SOH)** and reliable prediction of **Remaining Useful Life (RUL)** an open and critical challenge.

Traditional capacity-based and statistical methods are unable to capture the intricate spatio-temporal signatures embedded in high-frequency discharge waveforms. This project presents a **multi-modal deep learning framework** that jointly learns from two complementary data streams: high-frequency waveform data (Voltage, Current, dV/dt, Internal Resistance, Time) extracted from the critical CC-Discharge phase, and cycle-level auxiliary features (discharge energy, average voltage, previous SOH, global cycle index, discharge time, internal resistance).

A systematic architectural evolution spanning five implementations is reported, progressing from a baseline **CNN-LSTM** through a **Multi-Modal LSTM with Auxiliary Encoding**, a **Multi-Modal GRU**, a **Regularised GRU with Phase-Adaptive Hybrid Loss** (Cosine Similarity + Soft-DTW), and ultimately to an **Attentive Multi-Modal GRU with High-Resolution Spatial Encoding**. Each iteration introduced principled contributions: dual-stream feature processing, Batch Normalisation, curriculum training, and a learned Attention mechanism for intelligent noise filtering.

Evaluated on the publicly available **CALCE** dataset (CS2 and CX2 series), the final architecture achieves an R^2 of **0.9850** on the CS2_37 test cell across long-horizon (128-cycle) forecasting windows, with an MAE of 0.01614 and a MAPE of 3.108%.

In addition, a **BHAI (Battery Health Analysis Indicator)** web application was co-developed to translate the trained models into an accessible, production-grade health monitoring dashboard with real-time RUL prediction and one-click reporting.

2. Introduction

2.1 Background and Motivation

Rechargeable lithium-ion batteries are the dominant energy storage technology of the 21st century. From grid-scale solar storage to the drivetrain of an electric vehicle, their performance, safety, and economic viability hinge on accurate knowledge of the battery's current health state. Two quantities define this health state:

- **State of Health (SOH):** The ratio of a battery's current maximum usable capacity to its initial rated capacity, expressed as a percentage. A new battery has SOH $\approx$ 1.0 (100%); the industry end-of-life threshold is typically SOH = 0.80 (80%).
- **Remaining Useful Life (RUL):** The number of charge–discharge cycles (or calendar time) remaining before the battery's SOH falls below the end-of-life threshold.

Accurate SOH and RUL estimation are essential for reliable battery management systems (BMS), enabling optimal charge scheduling, early fault detection, warranty management, and safe second-life repurposing of retired cells.

2.2 Problem Statement

Battery degradation is a non-linear, stochastic process. Effective health estimation is hindered by both physical aging mechanisms and existing computational modeling gaps:

Physical Basis of Degradation

- **Irreversible Mechanisms:** Aging is driven by concurrent processes including Solid Electrolyte Interphase (SEI) growth, lithium plating, and active material cracking.
- **"Knee Point" Behavior:** Degradation exhibits a gradual initial decline followed by a critical "knee point," where capacity collapses rapidly and unpredictably.

Analytical & Observational Challenges

- **Underutilized Waveforms:** Traditional methods rely on simple capacity models, ignoring high-frequency discharge waveforms that contain critical aging signatures.
- **Spatio-Temporal Gaps:** Baseline models fail to concurrently capture short-term waveform patterns and long-term temporal trends across the battery lifecycle.
- **Data Defects:** BMS data is often corrupted by EMI, sensor jitter, and incomplete logs, creating "fuzzy" readings that mask true degradation trends.
- **Forecasting Drift:** Conventional models often diverge or "drift" when predicting health trajectories over long horizons exceeding 100 cycles.
- **Operational Sensitivity:** SOH is highly sensitive to environmental stressors like temperature and load, making robust generalization difficult.

2.3 Research Objectives

This project aims to:

1. Develop a multi-modal deep learning framework that integrates high-frequency discharge waveform features with cycle-level auxiliary signals.
2. Systematically improve and refine the proposed architecture through iterative experimentation and justified design enhancements.
3. Develop an appropriate loss function to ensure physically consistent and temporally accurate predictions.
4. Demonstrate strong generalisation across two distinct CALCE battery chemistries (CS2 and CX2 series).
5. Deploy the trained models on an interactive web application to demonstrate real-world usability and practical applicability.

3. Literature Review

3.1 Machine Learning Approaches

Support Vector Regression (SVR) and Gaussian Process Regression (GPR) [1] were applied to capacity fade prediction using hand-crafted features. While competitive on small datasets, these approaches require extensive feature engineering and often struggle to generalise across varying battery chemistries and operating conditions.

3.2 Deep Learning for Battery Prognostics

Recurrent architectures such as LSTM and GRU became prominent for time-series SOH modelling due to their ability to retain long-range dependencies. Zhang et al. [2] demonstrated LSTM-based SOH prediction on the CALCE dataset, reporting strong predictive performance with $R^2 > 0.97$. Convolutional Neural Networks (CNNs) have also been applied to extract spatial features from discharge curves [3], though often without integrating temporal dependencies.

3.3 Multi-Modal and Hybrid Architectures

Hybrid CNN-LSTM architectures [4] demonstrated that combining local waveform feature extraction with temporal sequence modelling improves prediction accuracy. However, the integration of high-frequency waveform information with auxiliary cycle-level features remains relatively underexplored in battery prognostics, motivating the proposed multi-modal framework in this work.

3.4 Loss Function Design for Time-Series Prediction

Traditional loss functions such as Mean Squared Error (MSE) optimise pointwise prediction accuracy but fail to capture temporal alignment and degradation trajectory consistency. Dynamic Time Warping (DTW) and its differentiable variants, such as Soft-DTW [5], have shown effectiveness in modelling non-linear temporal relationships and degradation patterns. These approaches are particularly useful for accurately learning the critical "knee" region in battery degradation trajectories.

4. Dataset

4.1 CALCE Battery Research Group

All experiments use data from the **CALCE (Center for Advanced Life Cycle Engineering) Battery Research Group** at the University of Maryland [6]. The dataset is publicly accessible at `https://calce.umd.edu/battery-data`.

4.2 Selected Cells and Experimental Conditions

Twelve cells in total were used to provide a robust training and evaluation corpus:

Table 1: CALCE Cell Series Summary

Series	**Cells Used**	**Chemistry**	**Nominal Capacity**
CS2	CS2_33 – CS2_38	Prismatic Lithium Cobalt Oxide (LCO)	1.1 Ah
CX2	CX2_33 – CX2_38	Structurally modified LCO	1.35 Ah

Experimental Conditions:

- **Temperature:** All tests performed at a constant 25°C to eliminate thermal variables.
- **Charging:** CC-CV protocol — Constant Current at 0.5C to 4.2 V, then Constant Voltage until current drops below 0.05 A.
- **Discharging:** Performed at 0.5C or 1C rates until a cut-off voltage of 2.7 V.

4.3 Data Structure and the Resetting Paradox

Raw data for each cell is distributed across multiple Excel workbooks, each capturing approximately 50 cycles. A critical subtlety—termed the **Resetting Paradox**—arises because:

- The `Cycle_Index` column resets to `1` at the start of each new file, while representing a *continuous* chronological aging process.
- `Charge_Capacity`, `Discharge_Capacity`, and `Energy` columns reset to `0.0` at each file boundary, acting as a per-file "odometer."

A custom **Global Synchronisation** pipeline was built to reconstruct the true continuous lifecycle by engineering a `global_cycle` counter and a `cell_id` field across all files.

4.4 The 7-Step Operational Cycle

The `Step_Index` column identifies distinct operational phases within each cycle. Understanding this structure is essential for isolating the discharge waveform used as CNN input. Figure 1 illustrates the full cycle.

Figure 1: The 7-Step battery operational cycle. Step 6 (CC Discharge) is the primary feature extraction window for the CNN spatial encoder.

4.5 Data Defects and Handling

Three categories of data defects were identified and addressed:

1. **Incomplete Cycles (Truncated Data):** Logging stops prematurely, producing artificial capacity drops. Addressed by filtering cycles with anomalously short discharge durations.
2. **Erroneous Readings and Outliers:** BMS communication errors produce "frozen" values or impossible spikes. Addressed with clipping and outlier removal (169 extreme outliers in `avg_voltage_discharge` were identified).
3. **Sensor Noise and EMI:** High-frequency jitter masks electrochemical signatures. Addressed by waveform normalisation and Batch Normalisation within the model.

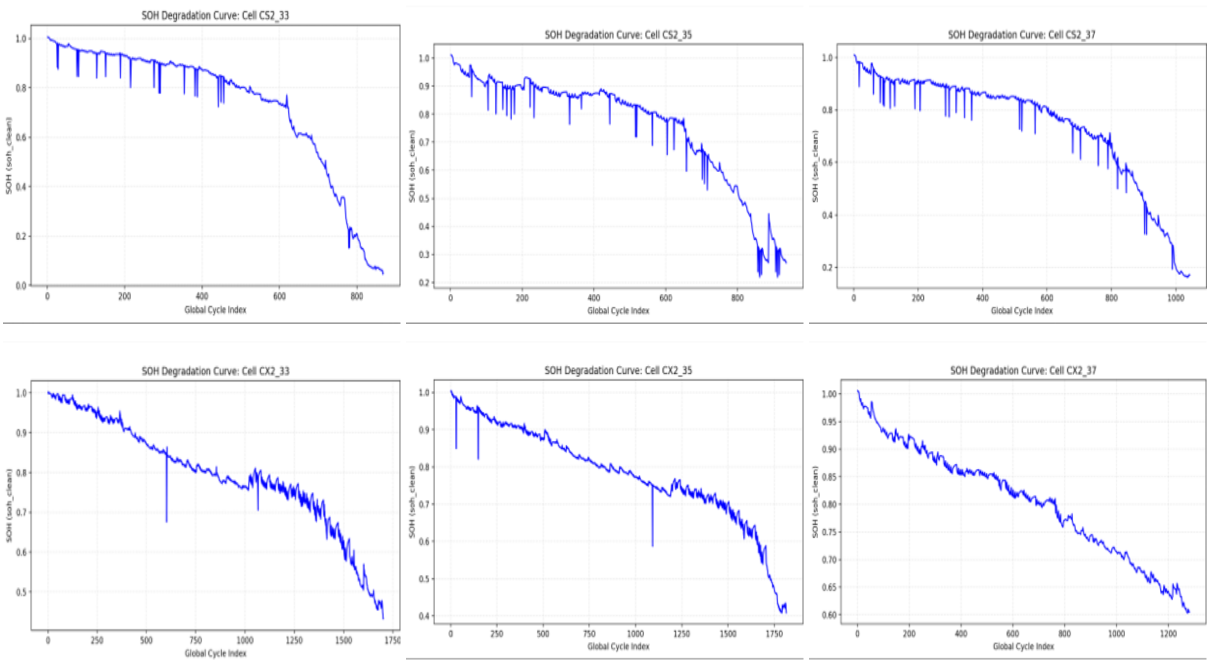


Figure 2: SOH Degradation Curves for CS2 and CX2 series highlighting inherent data defects and non-linear aging patterns[cite: 232, 242].

5. Problem Formulation

5.1 SOH Definition

Let C_{init} denote the *true initial capacity*, computed as the mean of the top-20 discharge capacity values within the first 30 cycles of a cell's life—a robust baseline that discards early conditioning artefacts. Then for cycle t:

$$\text{SOH}(t) = \frac{C_{\text{discharge}}(t)}{C_{\text{init}}} \tag{1}$$

where $C_{\text{discharge}}(t)$ is the delivered capacity at cycle t (in Ah).

5.2 Multi-Step Forecasting Formulation

Let $\mathbf{W}_t \in R^{L\times 5}$ denote the waveform tensor for cycle t (with L timesteps and 5 features), and $\mathbf{a}_t \in R^{d_a}$ the auxiliary feature vector. Given a sliding window of the past T_{in} cycles, the model learns a function:

$$f_\theta : (\{\mathbf{W}_{t-T_{\text{in}}+1}, \ldots, \mathbf{W}_t\},\ \{\mathbf{a}_{t-T_{\text{in}}+1}, \ldots, \mathbf{a}_t\}) \longmapsto \hat{\mathbf{s}}_{t+1:t+T_{\text{out}}} \tag{2}$$

where $\hat{\mathbf{s}}_{t+1:t+T_{\text{out}}} \in [0,1]^{T_{\text{out}}}$ is the predicted SOH trajectory over the next T_{out} cycles. Experiments use $(T_{\text{in}}, T_{\text{out}}) \in \{(30, 20),\ (128, 96),\ (256, 128)\}$.

5.3 Phase-Adaptive Hybrid Loss

Standard MSE loss ($\mathcal{L}_{\text{MSE}} = \frac{1}{T_{\text{out}}} \sum_i (\hat{s}_i - s_i)^2$) is blind to curve shape and timing. The hybrid loss alternates between two phases during training:

Phase A – Cosine-MSE (Trend Alignment):

$$\mathcal{L}_A = \mathcal{L}_{\text{MSE}} + \lambda_c \cdot (1 - \cos(\hat{\mathbf{s}}, \mathbf{s})) \tag{3}$$

This enforces global trend alignment—ensuring the predicted curve decays monotonically in the correct "direction."

Phase B – DTW-MSE (Timing Alignment):

$$\mathcal{L}_B = \mathcal{L}_{\text{MSE}} + \lambda_d \cdot \text{SoftDTW}(\hat{\mathbf{s}}, \mathbf{s}) \tag{4}$$

Soft-DTW [5] uses differentiable dynamic programming to penalise temporal misalignment, allowing the model to accurately locate the "knee point" even when it occurs earlier or later than the training average.

The full training objective alternates epochs between $\mathcal{L}_A$ and $\mathcal{L}_B$, forcing the model to simultaneously optimise global trend and local timing.

6. Methodology

6.1 Overall System Architecture

The proposed Spatial-Temporal Fusion pipeline is illustrated in Figure 3.

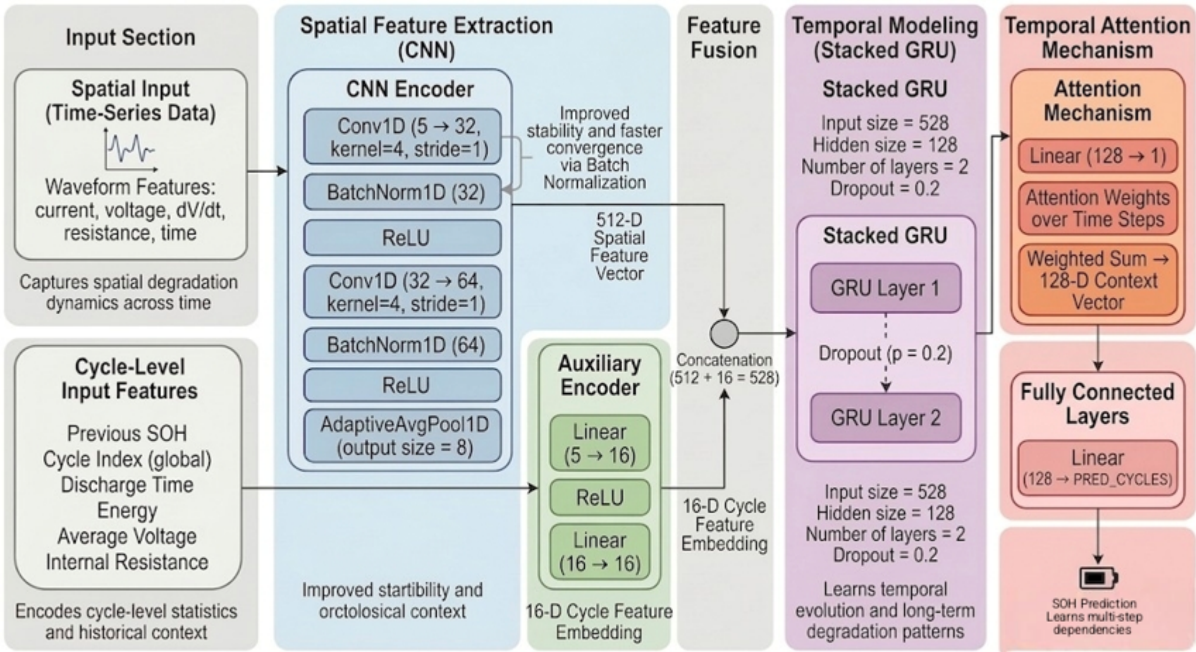


Figure 3: Spatial-Temporal Fusion pipeline of the 5th (final) implementation. Waveform and auxiliary feature streams are independently encoded, fused, passed through an Attention layer, and modelled temporally by a 2-layer GRU.

6.2 Data Processing and Feature Engineering

Format Conversion and Global Synchronisation. All CALCE Excel workbooks were converted to CSV format and merged using `global_cycle` and `cell_id` to reconstruct a continuous per-cell lifecycle.

Waveform Feature Extraction. Step 6 (CC Discharge) was isolated for each cycle. A `cycle_elapsed_time_sec` counter resets to $t = 0$ at the start of every discharge, providing a consistent temporal reference for the CNN.

Dual Scaling. Two independent `StandardScaler` instances were fit on training cells only:

- **Waveform Scaler:** Normalises (V, I, t, dV/dt, ir) to zero mean, unit variance.
- **Auxiliary Scaler:** Normalises cycle-level features and clips to $[-6, 6]$ to guard against gradient explosion from 169 extreme outliers in `avg_voltage_discharge`.

Cycle Padding. Discharge sequences are zero-padded to the maximum sequence length (`max_len`) to enable uniform tensor dimensions for batch processing.

Sliding Window Construction. A sliding window of width T_{in} with stride S is applied to generate (input, target) sample pairs. The target is the SOH vector for the next T_{out} cycles, clipped to $[0, 1.3]$ to exclude physically impossible values.

6.3 Architectural Evolution: Five Implementations

1st Implementation — CNN-LSTM with Sequential Feature Extraction. A baseline CNN extracts spatial features from the 5-channel discharge waveform. A single 128-unit LSTM processes the combined CNN output and auxiliary features over a 20-cycle window. The MLP prediction head maps the final LSTM hidden state to 20-cycle SOH forecasts.

2nd Implementation — Multi-Modal LSTM with Auxiliary Encoding and Robust Scaling. Core changes: (i) expanded auxiliary features to include discharge time, energy, average voltage, previous SOH, and global cycle index; (ii) introduced a dedicated auxiliary encoder (Sequential FC block) producing a 16-dimensional embedding; (iii) increased LSTM input dimensionality ($\approx$67 $\rightarrow$ 80); (iv) applied dual scalers and outlier clipping.

3rd Implementation — Multi-Modal GRU. The LSTM layer was replaced with a GRU, which has a simpler gating mechanism with fewer parameters. Empirically, GRU provided faster convergence and superior per-cycle accuracy, particularly on the CX2 series.

4th Implementation — Regularised Multi-Modal GRU with Phase-Adaptive Hybrid Loss. Three regularisation techniques were added: **BatchNorm1d** after convolutional layers (stabilise activations), **GRU Dropout** (20%, reduce temporal overfitting), and **FC Dropout** (20%, expanded head 128 $\rightarrow$ 256 $\rightarrow$ T_{out}). Training employed the **Phase-Adaptive Hybrid Loss** (alternating Cosine-MSE and Soft-DTW-MSE epochs) as defined in Section 5.3.

5th Implementation — Attentive Multi-Modal GRU with High-Resolution Spatial Encoding. Key architectural innovations:

- **High-Resolution Spatial Encoding:** `AdaptiveAvgPool1d(8)` retains 8 spatial points (versus a single global mean), expanding the CNN output to 512 dimensions. This preserves the waveform "topography" including the shifting knee region.
- **Learned Attention Mechanism:** A global softmax attention head computes cycle-level importance weights over the T_{in} input window. Noisy or uninformative cycles receive low weight; cycles exhibiting true degradation signatures are amplified.
- **Expanded GRU Capacity:** Hidden units doubled from 128 to 256 to track complex interaction patterns over longer sequences.
- **Streamlined Prediction Head:** Direct mapping GRU hidden state $\rightarrow$ T_{out} removes intermediate layers, reducing parameter redundancy and overfitting.

7. Results and Discussion

7.1 Evaluation Protocol and Metrics

All models are evaluated on held-out test cells (CS2_37 and CX2_37) not seen during training or validation. Performance is measured by:

- R^2 Score (coefficient of determination): higher is better; 1.0 = perfect.
- MAE (Mean Absolute Error) on SOH units: lower is better.
- RMSE (Root Mean Square Error): lower is better; more sensitive to large errors.
- MAPE (%) (Mean Absolute Percentage Error): lower is better; scale-independent.
- Max Error: lower is better, indicating the worst-case prediction boundary.
- Correlation (Pearson): higher is better (closer to 1.0); measures the linear relationship and how well the predicted trend tracks the true degradation curve.

7.2 Implementation Comparison: Short-Horizon Window (30-in / 20-out)

Table 2: Model comparison on **CS2_37** test cell (Window: 30 in / 20 out)

Implementation	R^2	**MAE**	**RMSE**	**MAPE (%)**	**Max Error**	**Corr.**
1st: CNN-LSTM	0.9863	0.01543	0.02466	3.167	0.15160	0.9938
2nd: Multi-Modal LSTM + Encoding	0.9884	0.01432	0.02261	2.654	0.12816	0.9946
3rd: Multi-Modal GRU	0.9899	0.01246	0.02107	2.244	0.12627	0.9950
4th: Regularised GRU + Hybrid Loss	0.9878	0.01320	0.02242	2.456	0.13316	0.9940
5th: Attentive GRU + Hi-Res Spatial	0.9848	0.01497	0.02503	3.224	0.13875	0.9935

Table 3: Model comparison on **CX2_37** test cell (Window: 30 in / 20 out)

Implementation	R^2	**MAE**	**RMSE**	**MAPE (%)**	**Max Error**	**Corr.**
1st: CNN-LSTM	0.9634	0.01568	0.01855	1.960	0.05640	0.9942
2nd: Multi-Modal LSTM + Encoding	0.9780	0.01176	0.01424	1.487	0.03578	0.9948
3rd: Multi-Modal GRU	0.9908	0.00720	0.00922	0.914	0.02927	0.9962
4th: Regularised GRU + Hybrid Loss	0.9918	0.00687	0.00859	0.868	0.03194	0.9959
5th: Attentive GRU + Hi-Res Spatial	0.9691	0.01403	0.01667	1.737	0.04080	0.9953

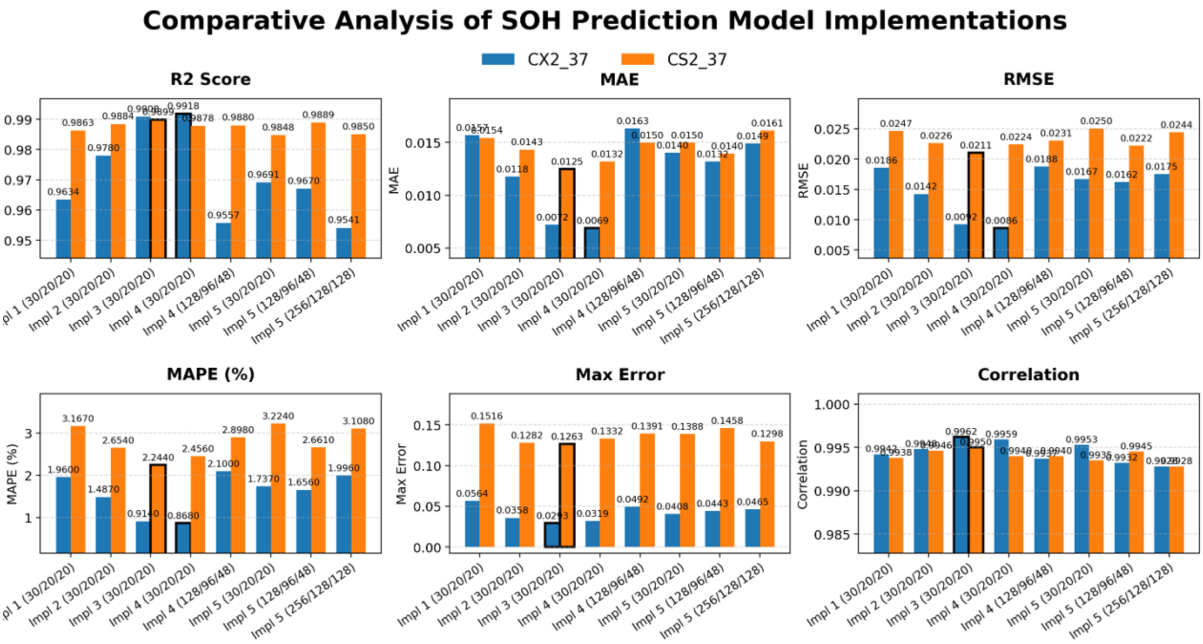
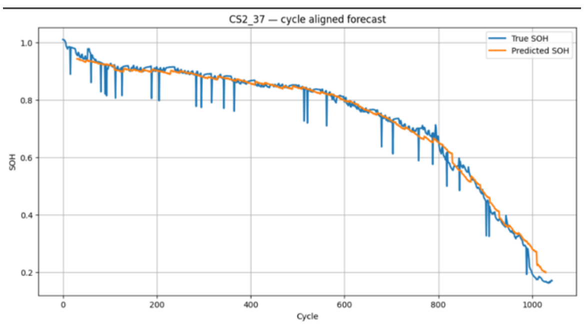


Figure 4: Comparative Analysis of SOH Prediction Model Implementations across key performance metrics.

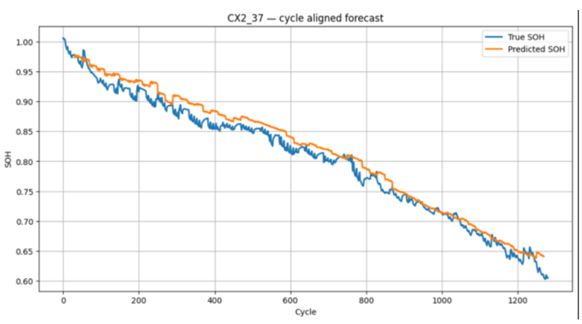
7.3 Long-Horizon Forecasting Results

Table 4: 5th Implementation on CS2_37 and CX2_37 across three prediction horizons

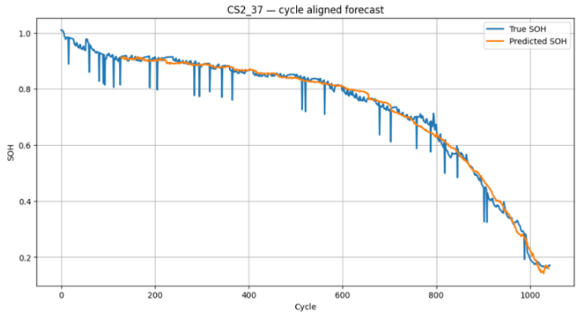
Window Config	Cell	R^2	MAE	RMSE	MAPE (%)	Max Error	Corr.
30-in / 20-out	CS2_37	0.9848	0.01497	0.02503	3.224	0.13875	0.9935
	CX2_37	0.9691	0.01403	0.01667	1.737	0.04080	0.9953
128-in / 96-out	CS2_37	0.9889	0.01395	0.02222	2.661	0.14578	0.9945
	CX2_37	0.9670	0.01320	0.01619	1.656	0.04434	0.9932
256-in / 128-out	CS2_37	0.9850	0.01614	0.02445	3.108	0.12980	0.9928
	CX2_37	0.9541	0.01490	0.01753	1.996	0.04648	0.9928



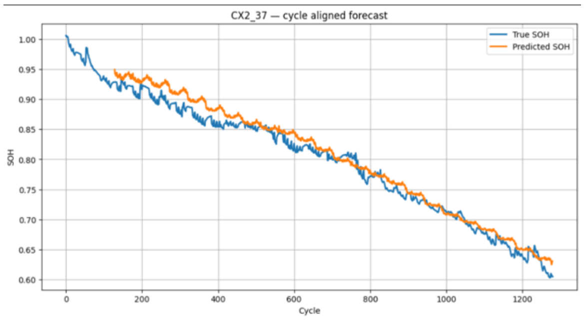
(a) CS2_37 (30-in / 20-out)



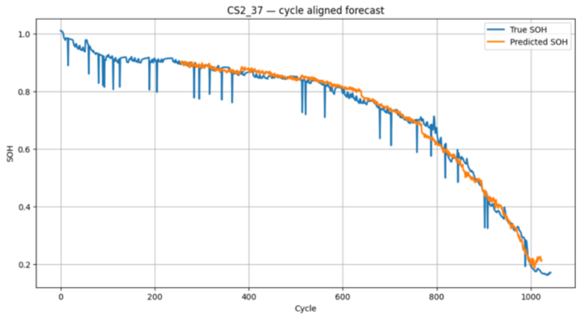
(b) CX2_37 (30-in / 20-out)



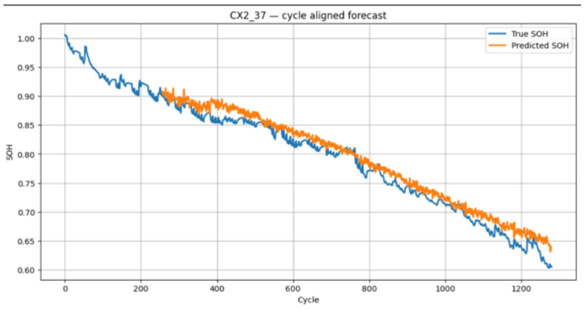
(c) CS2_37 (128-in / 96-out)



(d) CX2_37 (128-in / 96-out)



(e) CS2_37 (256-in / 128-out)



(f) CX2_37 (256-in / 128-out)

Figure 5: 5th Implementation: SOH prediction trajectories across multiple forecasting horizons for CS2 and CX2 test cells.

7.4 Key Findings and Analysis

GRU outperforms LSTM. Replacing LSTM with GRU (3rd implementation) produced a significant jump on CX2_37, improving R^2 from 0.9780 to 0.9908 and halving the MAE. GRU's simpler gating appears to generalise better with the available data volume.

Phase-Adaptive Hybrid Loss improves long-horizon stability. The 4th implementation's curriculum training strategy showed its most pronounced benefits on longer prediction windows (96 cycles), where previous MSE-only models exhibited "drift." The Cosine phase keeps global trends physically realistic; the DTW phase accurately localises end-of-life capacity cliffs.

High-Resolution Spatial Encoding and Attention improve curve fidelity. The 5th implementation's `AdaptiveAvgPool1d(8)` encoder preserves the discharge waveform shape that a global mean pooling destroys. The Attention mechanism suppresses noisy or uninformative cycles, producing smoother and more physically plausible SOH trajectories without oscillations.

Long-horizon forecast stability. The system successfully performs up to 128-cycle ahead forecasting (window 256-in / 128-out) with $R^2 > 0.95$ on CX2_37 and $R^2 = 0.985$ on CS2_37, demonstrating practical RUL prediction capability.

Cross-cell generalisation. Consistent performance across both the CS2 and CX2 series—despite their different aging trajectories and "knee" locations—validates the framework's ability to learn transferable degradation representations.

8. BHAI: Battery Health Analysis Indicator

8.1 Overview

The **BHAI (Battery Health Analysis Indicator)** web application is an interactive dashboard that operationalises the predictive models developed in this project. Its purpose is to bridge the gap between research-grade deep learning and the practical needs of battery engineers, fleet operators, and BMS developers.

- **Backend:** Powered by the CNN-LSTM and CNN-GRU-Attention models trained in this project.
- **Impact:** Instantly translates raw sensor CSV data into actionable RUL predictions with statistical confidence bounds ($\pm 1\sigma$).

8.2 Core Platform Capabilities

Seamless Interactive Dashboard. An intuitive drag-and-drop CSV upload interface with automated data normalisation prepares raw sensor data for immediate model inference, requiring no user preprocessing.

Multi-Modal Architecture Selection. Users can flexibly switch between the CNN-LSTM and CNN-GRU-Attention backends, supporting diverse dataset domains including experimental cycling data and BMS telemetry.

Dynamic Parameter Control. Customisable failure thresholds (e.g., SOH = 0.80) and adjustable prediction window configurations (e.g., 64-in / 48-out) allow tailoring to specific operational requirements.

8.3 Analytical Engine and Reporting

Rich Visualisations. Interactive forecast charts display SOH trajectories with $\pm 1\sigma$ confidence bands and end-of-life threshold markers. A "Raw Data" tab enables cycle-by-cycle auditing.

Intelligent Early Warning System. A dynamic status banner alerts users when the battery is projected to cross the failure threshold within the current forecast window.

Actionable KPI Metrics. Automatically computed health markers include: Last Known SOH, Remaining Useful Life (in cycles), and the cycle-over-cycle Degradation Rate.

Automated Reporting and Export. One-click export produces full predicted trajectory CSVs and formatted text summary reports for offline analysis and documentation.

Future Diagnostic Insights. Planned functionality includes identification of specific degradation mechanisms (SEI Growth, Lithium Plating) and targeted prevention strategy recommendations.

9. Conclusion and Future Work

9.1 Summary of Contributions and Novelty

This project developed a multi-modal deep learning framework for battery health estimation, achieving the following key milestones and innovations:

- **Architectural Evolution:** Executed a systematic 5-stage evolution from baseline CNN-LSTM to an Attentive Multi-Modal GRU, demonstrating that GRU structures offer superior stability and accuracy for battery time-series data.
- **Multi-Modal Fusion:** Developed a dual-stream architecture that extracts spatial features from raw waveforms while simultaneously processing cycle-level metadata , providing a more holistic health assessment.
- **Adaptive Hybrid Loss:** Introduced a curriculum training strategy alternating between Cosine-MSE (for global trend alignment) and Soft-DTW-MSE (for precise timing of capacity "cliffs"), significantly reducing long-horizon forecasting drift.
- **High-Resolution Encoding:** Implemented 8-point adaptive spatial pooling to preserve electrochemical signatures within the discharge curve, integrated with a learned Attention mechanism to filter sensor noise.
- **Technical Performance:** Achieved a high accuracy of $R^2 \approx 0.98$ on held-out test cells, proving the model's ability to generalize across different battery chemistries and aging profiles.
- **Operational Deployment:** Co-developed and deployed the **BHAI (Battery Health Analysis Indicator)** web application, a production-ready dashboard that provides real-time SOH monitoring and automated health reporting.

9.2 Limitations

- Dataset limited to CALCE cells at 25°C; temperature variability and real-world drive-cycle patterns are not represented.
- The Soft-DTW computation is memory-intensive; scaling to very long sequences ($T_{\text{out}} > 200$) may require approximate DTW implementations.
- No prospective validation has been conducted; all results are retrospective on historical cycling data.
- The five-step cycle structure is specific to CALCE experimental conditions; adaptation to BMS-sourced telemetry with different step conventions requires preprocessing redesign.

9.3 Future Directions

1. **Temperature-Aware Multi-Condition Training:** Incorporate thermal data and variable rate experiments (NASA, Oxford datasets) to build a temperature-conditioned model.
2. **Transformer-Based Architecture:** Explore Temporal Fusion Transformer (TFT) as an alternative to GRU, leveraging multi-head self-attention for long-range dependency modelling.
3. **Physics-Informed Learning:** Incorporate degradation physics constraints (monotonic capacity decline, positive RMSE growth) as soft penalties in the loss function.
4. **Prospective Clinical/Industrial Validation:** Partner with an EV manufacturer or grid storage operator to validate the framework on real-world BMS telemetry under operational conditions.
5. **BHAI Diagnostic Module:** Extend the web app with automated degradation mode identification (SEI Growth, Lithium Plating) based on voltage and dV/dt signature analysis.

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